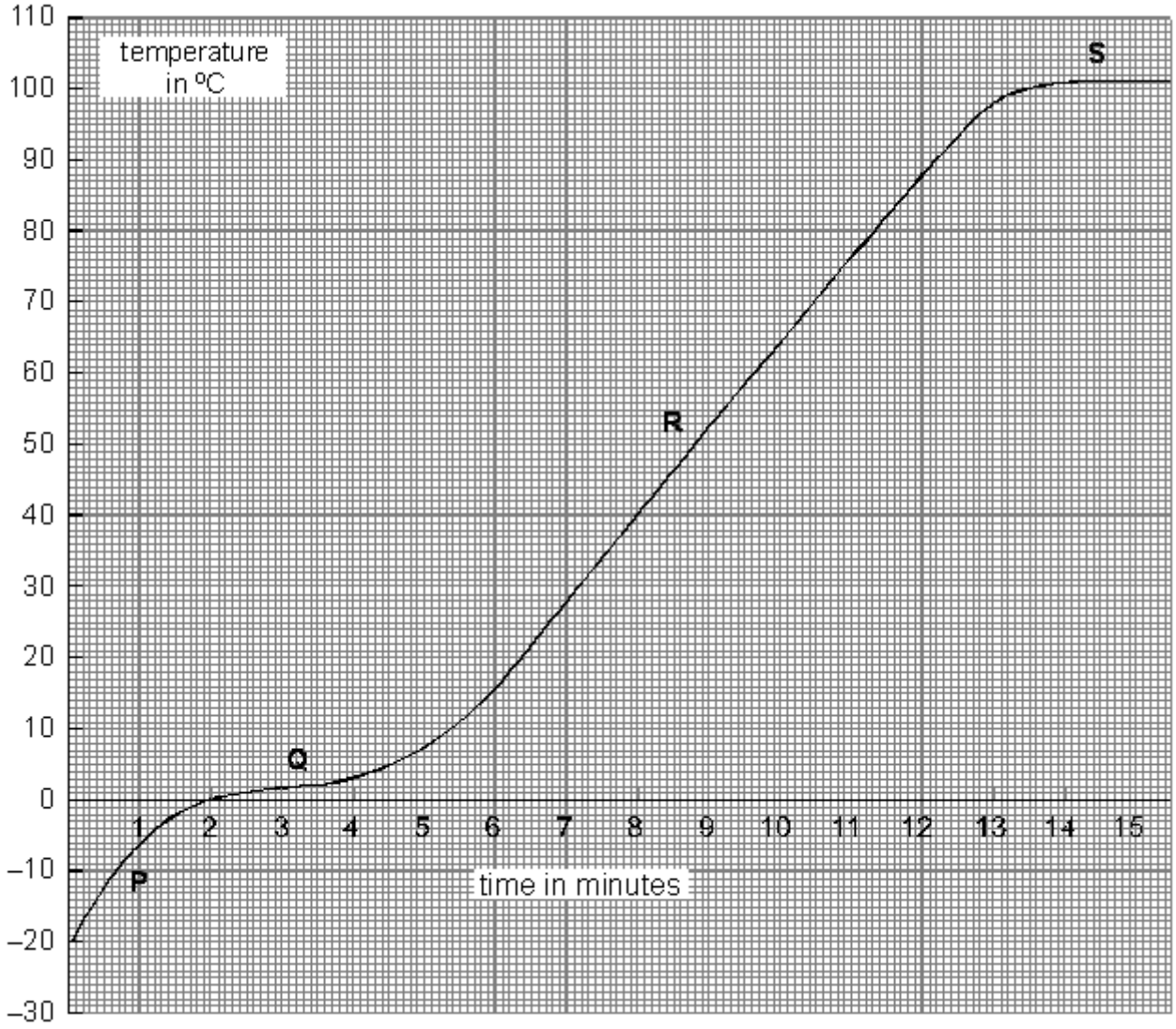
Q1.

Some crushed ice at –20° C was placed in a beaker. A thermometer was put into the ice, and the beaker was heated gently for 15 minutes.

The graph shows how the reading on the thermometer changed over the 15 minutes.



(a) By how much did the temperature in the beaker change during the 15 minutes?

..... °C

1 mark

(b) Which letter on the graph shows:

(i) when the ice is melting? .....

1 mark

(ii) when the water is boiling?.....

1 mark

(c) During the experiment, the beaker and its contents were quickly removed from the heat and weighed on a balance at the following times.

at 0 minutes

at 5 minutes

at 10 minutes

at 15 minutes

(i) At which **two** times would you expect the readings on the balance to be the same?

..... minutes and ..... minutes

1 mark

(ii) Between which **two** of these times was the mass of the contents of the beaker changing most rapidly?

..... minutes and ..... minutes

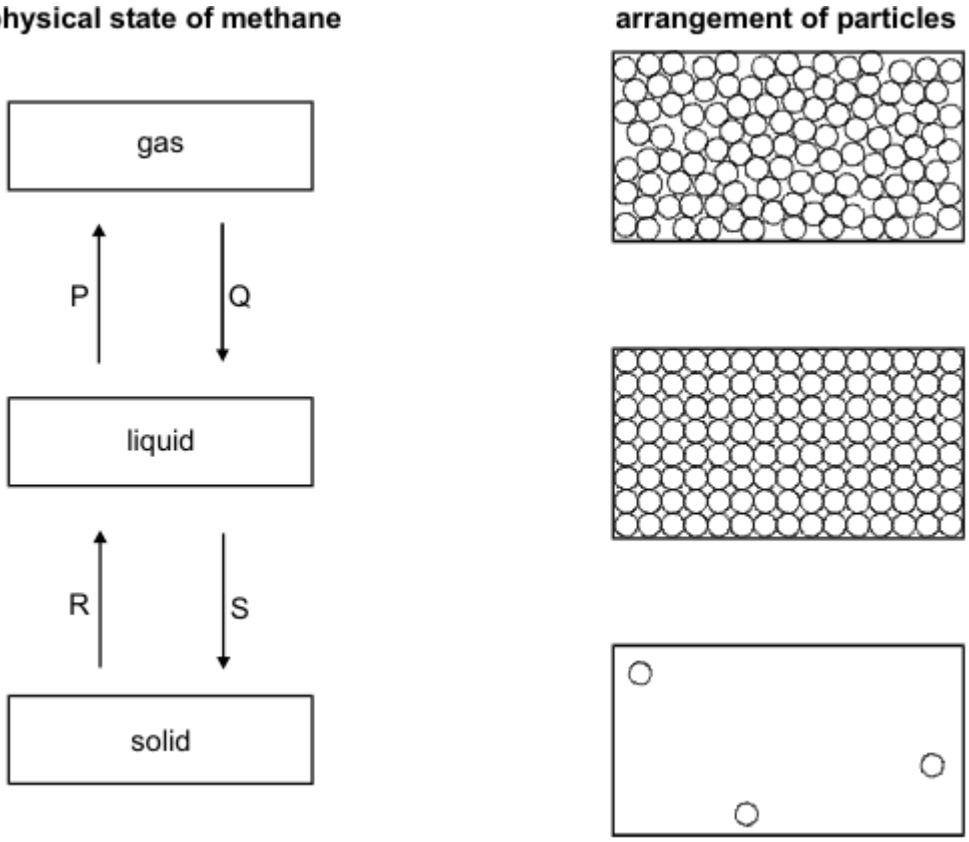
1 mark

Maximum 5 marks

Q2.

(a) Methane can be a gas, a liquid or a solid. In the diagram below, arrows P, Q, R and S represent changes of state.

The boxes on the right show the arrangement of particles of methane in the three different physical states.
Each circle represents a particle of methane.



(i) Draw a line from each physical state of methane to the arrangement of particles in that physical state.
Draw only **three** lines.

1 mark

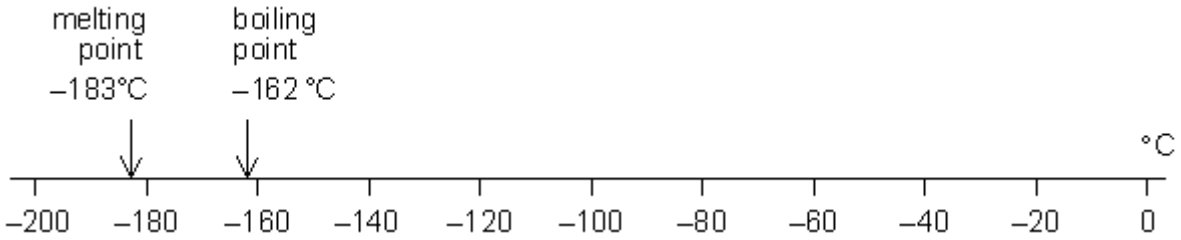
(ii) Arrows P, Q, R and S represent changes of state.
Which arrow represents:

evaporation? .....

melting? .....

2 marks

(b) Methane is the main compound in natural gas. The scale below shows the melting point and the boiling point of methane.



Methane has three physical states: solid, liquid and gas.

(i) What is the physical state of methane at –170°C?

.....

1 mark

(ii) The formula of methane is CH_4. The symbols for the two elements in methane are C and H.

Give the names of these two elements.

element C .....

element H .....

2 marks

(iii) When methane burns, it reacts with oxygen.
One of the products is water, H_2O.

Give the name of the other product.

.....

1 mark
Maximum 7 marks

Q1.

(a) 121

*do **not** accept '120'*

1 (L5)

(b) (i) Q

1 (L5)

(ii) S

1 (L5)

(c) (i) 0 and 5

***both** stages are required for the mark*

1 (L5)

(ii) 10 and 15

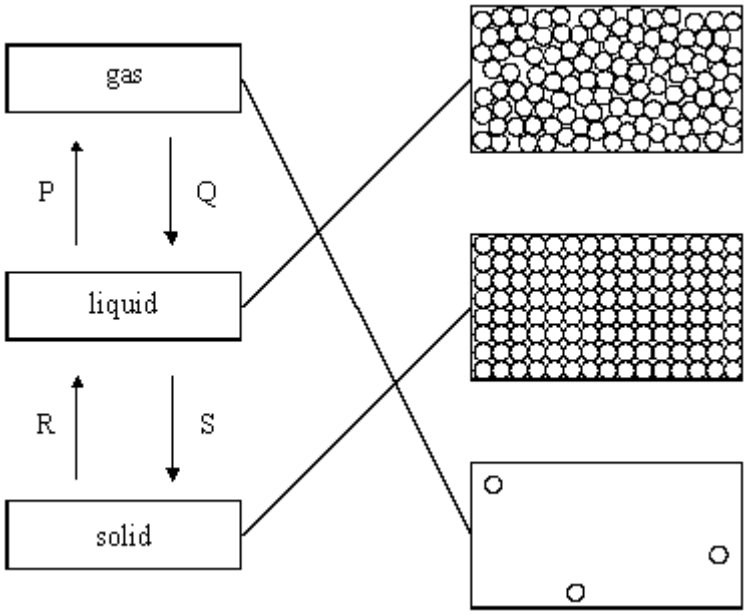
***both** stages are required for the mark*

1 (L5)

[5]

Q2.

(a) (i)



***all three** lines must be correct for the mark*

1 (L6)

(ii) *evaporation*: P

1 (L5)

melting: R

1 (L5)

(b) (i) liquid

1 (L6)

(ii) carbon

1 (L6)

hydrogen

1 (L6)

(iii) carbon dioxide

accept 'CO_2'

accept 'carbon monoxide' ***or*** *'CO'*

accept 'carbon' ***or*** *'soot'*

answers must be in the correct order

1 (L6)

[7]